For the use only of a Registered Medical Practitioner or a Hospital or a Laboratory

Arimac 5 / 10

Donepezil Hydrochloride Tablets 5 mg / 10 mg

COMPOSITION

Each film coated tablet contains:
Donepezil hydrochloride …………5 mg / 10mg

PRODUCT DESCRIPTION

Arimac 5 : White to off – white, round shaped, biconvex, film coated tablets debossed with "ML 89" on one side and plain on the other side.

Arimac 10 : Yellow colour, round shaped, biconvex, film – coated tablet debossed "ML 88" on one side and plain on the other side.

PHARMACODYNAMIC EFFECT

Pharmacodynamics

ATC Code for Donepezil : N06DA02
Pharmacotherapeutic group : Acetylcholinesterase inhibitor

Mode of Action

Donepezil hydrochloride is a specific and reversible inhibitor of acetylcholinesterase, the redominant cholinesterase in the brain. Donepezil hydrochloride is in vitro over 1000 times more potent an inhibitor of this enzyme than of butyrylcholinesterase, an enzyme that is present mainly outside the central nervous system.

Pharmacokinetics:

Absorption: Maximum plasma levels are reached approximately 3 to 4 hours after oral administration. Plasma concentrations and area under the curve rise in proportion to the dose. The terminal disposition half-life is approximately 70 hours, thus, administration of multiple single-daily doses results in gradual approach to steady-state. Approximate steady-state is achieved within 3 weeks after initiation of therapy. Food does not influence the rate or extent of absorption of donepezil tablets.

Distribution: Donepezil hydrochloride is approximately 95% bound to human plasma proteins. The plasma protein binding of the active metabolite 6-O-desmethyldonepezil is not known. The distribution of donepezil hydrochloride in various body tissues has not been definitively studied.

Metabolism & Excretion:

Donepezil hydrochloride is both excreted in the urine intact and metabolised by the cytochrome P450 system to multiple metabolites, not all of which have been identified. Following administration of a single 5 mg dose of 14C-labeled donepezil hydrochloride, plasma radioactivity, expressed as a percent of the administered dose, was present primarily as intact donepezil hydrochloride (30%), 6-O-desmethyl donepezil (11% - only metabolite that exhibits activity similar to donepezil hydrochloride), donepezil-cis-N-oxide (9%), 5-O-desmethyl donepezil (7%) and the glucuronide conjugate of 5-O-desmethyl donepezil (3%). Approximately 57% of the total administered radioactivity was recovered from the urine (17% as unchanged donepezil), and 14.5% was recovered from the faeces, suggesting biotransformation and urinary excretion as the primary routes of elimination. There is no evidence to suggest enterohepatic recirculation of donepezil hydrochloride and/or any of its metabolites. Plasma donepezil concentrations decline with a half-life of approximately 70 hours.

INDICATIONS AND USAGE:

Arimac Tablets are indicated for the symptomatic treatment of mild , moderate & severe dementia in Alzheimer's disease.

CONTRAINDICATIONS:

Donepezil is contraindicated in patients with a known hypersensitivity to donepezil hydrochloride, piperidine derivatives, or to any excipients used in the formulation.

ADVERSE REACTION:

The most common adverse events are diarrhoea, muscle cramps, fatigue, nausea, vomiting and insomnia.
The adverse effects are as follows :

System Organ Class	Very Common	Common	Uncommon	Rare	Very Rare	Not Known
Infections and infestations		Common cold				
Metabolism and nutrition disorders		Anorexia				
Psychiatric disorders		Hallucinations** Agitation** Aggressive behaviour** Abnormal dreams and nightmares**				
Nervous system disorders		Syncope* Dizziness Insomnia	Seizure*	Extrapyramidal symptoms	Neuroleptic malignant syndrome	
Cardiac disorders			Bradycardia	Sino-atrial block Atrioventricular block		Polymorphic ventricular tachycardia including Torsade de Pointes; Electrocardiogram QT interval prolonged.
Gastrointestinal disorders	Diarrhoea Nausea	Vomiting Abdominal disturbance	Gastrointestinal haemorrhage Gastric and duodenal ulcers Salivary hypersecretion			
Hepato-biliary disorders				Liver dysfunction including hepatitis***		
Skin and subcutaneous tissue disorders		Rash Pruritus				
Musculoskeletal, connective tissue and bone disorders		Muscle cramps			Rhabdomyolysis****	
Renal and urinary disorders		Urinary incontinence				
General disorders and administration site conditions	Headache	Fatigue Pain				
Investigations			Minor increase in serum concentration of muscle creatine kinase			
Injury and poisoning		Accident				

*In investigating patients for syncope or seizure the possibility of heart block or long sinusal pauses should be considered
**Reports of hallucinations, abnormal dreams, nightmares, agitation and aggressive behaviour have resolved on dose-reduction or discontinuation of treatment.
***In cases of unexplained liver dysfunction, withdrawal of Arimac should be considered.
****Rhabdomyolysis has been reported to occur independently of Neuroleptic Malignant syndrome and in close temporal association with donepezil initiation or dose increase.

DOSAGE

Adults/Elderly:

Mild to Moderate Alzheimer's Disease:

Treatment is initiated at 5 mg/day (once-a-day dosing). Donepezil should be taken orally, in the evening, just prior to retiring. Donepezil can be taken with or without food and should be swallowed whole with water. The 5 mg/day dose should be maintained for at least one month in order to allow the earliest clinical responses to treatment to be assessed and to allow steady-state concentrations of donepezil hydrochloride to be achieved. Following a one-month clinical assessment of treatment at 5 mg/day, the dose of Donepezil can be increased to 10 mg/day (once-a-day dosing). The maximum recommended dose is 10 mg.

Severe Alzheimer's Disease:
Treatment should be initiated and supervised by a physician experienced in the diagnosis and treatment of Alzheimer's dementia. Diagnosis should be made according to accepted guidelines (e.g. DSM IV, ICD 10). Therapy with donepezil should only be started if a treatment can be continued for as long as a therapeutic benefit for the patient exists. Therefore, the clinical benefit of donepezil should be reassessed on a regular basis. Discontinuation should be considered when evidence of a therapeutic effect is no longer present. Individual response to donepezil cannot be predicted. Upon discontinuation of treatment, a gradual abatement of the beneficial effects of Donepezil is seen.

Renal and hepatic impairment:
A similar dose schedule can be followed for patients with renal impairment, as clearance of donepezil hydrochloride is not affected by this condition.
Due to possible increase exposure in mild to moderate hepatic impairment, dose escalation should be performed according to individual tolerability.
There are no data for patients with severe hepatic impairment.

Children:
Donepezil is not recommended for use in children.

DRUG INTERACTIONS:
Donepezil hydrochloride and/or any of its metabolites do not inhibit the metabolism of theophylline, warfarin, cimetidine or digoxin in humans. The metabolism of donepezil hydrochloride is not affected by concurrent administration of digoxin or cimetidine.
Enzyme inducers, such as rifampicin, phenytoin, carbamazepine and alcohol may reduce the levels of donepezil. Since the magnitude of an inhibiting or inducing effect is unknown, such drug combinations should be used with care. Donepezil hydrochloride has the potential to interfere with medications having anticholinergic activity. There is also the potential for synergistic activity with concomitant treatment involving medications such as succinylcholine, other neuro-muscular blocking agents or cholinergic agonists or beta blocking agents that have effects on cardiac conduction.

Cases of QT interval prolongation and Torsade de Pointes have been reported for donepezil. Caution is advised when donepezil is used in combination with other medicinal products known to prolong the QT interval and clinical monitoring (ECG) may be required. Examples include:

- Class IA antiarrhythmics (e.g. quinidine).
- Class III antiarrhythmics (e.g. amiodarone, sotalol).
- Certain antidepressants (e.g. citalopram, escitalopram, amitriptyline).
- Other antipsychotics (e.g. phenothiazine derivatives, sertindole, pimozide, ziprasidone).
- Certain antibiotics (e.g. clarithromycin, erythromycin, levofloxacin, moxifloxacin).

WARNINGS & PRECAUTIONS:
a) The administration of Arimac concomitantly with other inhibitors of acetylcholinesterase, agonists or antagonists of the cholinergic system should be avoided.
b) Severe Hepatic Impairment: There are no data for patients with severe hepatic impairment.
Anaesthesia: Donepezil, as a cholinesterase inhibitor, is likely to exaggerate succinylcholine-type muscle relaxation during anaesthesia.
Cardiovascular Conditions: There have been post-marketing reports of QT interval prolongation and Torsade de Pointes. Caution is advised in patients with pre-existing or family history of QT prolongation, in patients treated with drugs affecting the QT interval, or in patients with relevant pre-existing cardiac disease (e.g. uncompensated heart failure, recent myocardial infarction, bradyarrhythmias), or electrolyte disturbances (hypokalaemia, hypomagnesaemia). Clinical monitoring (ECG) may be required.
Gastrointestinal Conditions: . Patients at increased risk for developing ulcers, e.g., those with a history of ulcer disease or those receiving concurrent nonsteroidal anti inflammatory drugs (NSAIDs), should be monitored for symptoms.However, the studies with donepezil hydrochloride showed no increase, relative to placebo, in the incidence of either peptic ulcer disease or gastrointestinal bleeding.
Genitourinary: Cholinomimetics may cause bladder outflow obstruction.
Neurological Conditions: Seizures: Cholinomimetics are believed to have some potential to cause generalized convulsions. However, seizure activity may also be a manifestation of Alzheimer's Disease.
Pulmonary Conditions: Because of their cholinomimetic actions, cholinesterase inhibitors should be prescribed with care to patients with a history of asthma or obstructive pulmonary disease.

Effects on ability to drive and use machines
Donepezil has minor or moderate influence on the ability to drive and use machines. Dementia may cause impairment of driving performance or compromise the ability to use machinery. Furthermore, donepezil can induce fatigue, dizziness and muscle cramps, mainly when initiating or increasing the dose. The treating physician should routinely evaluate the ability of patients on donepezil to continue driving or operating complex machines.

PREGNANCY AND LACTATION:
Pregnancy
There are no adequate data from the use of donepezil in pregnant women. The potential risk for humans is unknown. Donepezil should not be used during pregnancy unless clearly necessary. It should be used during pregnancy only if the potential benefit justifies the potential risk to the fetus.
Lactation :
It is not known whether donepezil hydrochloride is excreted in human breast milk and there are no studies in lactating women. Donepezil hydrochloride should only be used by a woman who is breastfeeding if the potential benefit outweighs the potential risk to the infant.

OVERDOSAGE:
The estimated median lethal dose of donepezil hydrochloride following administration of a single oral dose in mice and rats is 45 and 32 mg/kg, respectively, or approximately 225 and 160 times the maximum recommended human dose of 10 mg per day. Dose-related signs of cholinergic stimulation were observed in animals and included reduced spontaneous movement, prone position, staggering gait, lacrimation, clonic convulsions, depressed respiration, salivation, miosis, fasciculation and lower body surface temperature.
Overdosage with cholinesterase inhibitors can result in cholinergic crisis characterized by severe nausea, vomiting, salivation, sweating, bradycardia, hypotension, respiratory depression, collapse and convulsions. Increasing muscle weakness is a possibility and may result in death if respiratory muscles are involved.
As in any case of overdose, general supportive measures should be utilised. Tertiary anticholinergics such as atropine may be used as an antidote for Donepezil overdosage. Intravenous atropine sulphate titrated to effect is recommended: an initial dose of 1.0 to 2.0 mg IV with subsequent doses based upon clinical response. Atypical responses in blood pressure and heart rate have been reported with other cholinomimetics when co-administered with quaternary anticholinergics such as glycopyrrolate. It is not known whether donepezil and/or its metabolites can be removed by dialysis (haemodialysis, peritoneal dialysis, or hemofiltration).

STORAGE :
Store below 30°C in a dry place and protect from light.
KEEP OUT OF REACH OF CHILDREN.

PRESENTATION :
Alu / Clear PVC / PVdC blister pack of 10 Tablets. Such 3 blisters packed in a carton along with pack insert.

For further information please consult your physician or pharmacist.

MANUFACTURED BY :
MACLEODS PHARMACEUTICALS LTD.
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